Sime Darby Township Hackathon · Theme: Planet

SunD

Powering every drive with Sime Darby sunshine.

SunD Wallet + **SunD Bay**: CREAM rooftops earn the rewards, and SunD Bays' own canopies power cheaper, cleaner rides for everyone, in Elmina and beyond.

Energy efficiency · Green transport · Circular economy



Team **UrbanByte**

SunD

EVs are outrunning chargers, and those chargers run on fossil power

19 : 1

EVs per public charger

vs

122,286 EVs vs 6,416 chargers · 10,000-charger target for 2025 missed

92%

of Peninsular power is fossil

Coal 58.6% · Gas 33.1% · Other 8.3%

27%

of would-be buyers held back by too few chargers

- 61% finding one
- 53% waiting
- 38% cost

INSIDE SIME DARBY TOWNSHIPS

CREAM cash leaks out of the loop

Rooftop lease pays cash, spent anywhere. Residents never use their own solar.

Shared solar is hard to deliver

5 km cap · 9 sen/kWh access charge · midday surplus, night shortfall.

Who is affected EV drivers and would-be buyers · SD residents · Sime Darby · the planet

Why now 30,000-charger target by 2030 · SD net zero by 2050 · CREAM just launched

Sources: MITI (May 2026), carz.com.my, EY Malaysia EV survey, Reuters/The Edge power mix 2025

SunD

Five groups, one gap: solar people can use



SD landed owners

- Roof solar they never use
- **Solar income**: credits +20% or bill rebate



SD condo residents

- No charger space, little roof
- **Canopy bays** in their car park, 15% off



Public EV drivers

- Few bays, long waits, pricey DC
- **Cheaper rates**, surplus deals, Saver +50%



Future EV buyers

- Charging anxiety blocks the buy
- **Bays nearby** + low running cost → go EV



Sime Darby Property

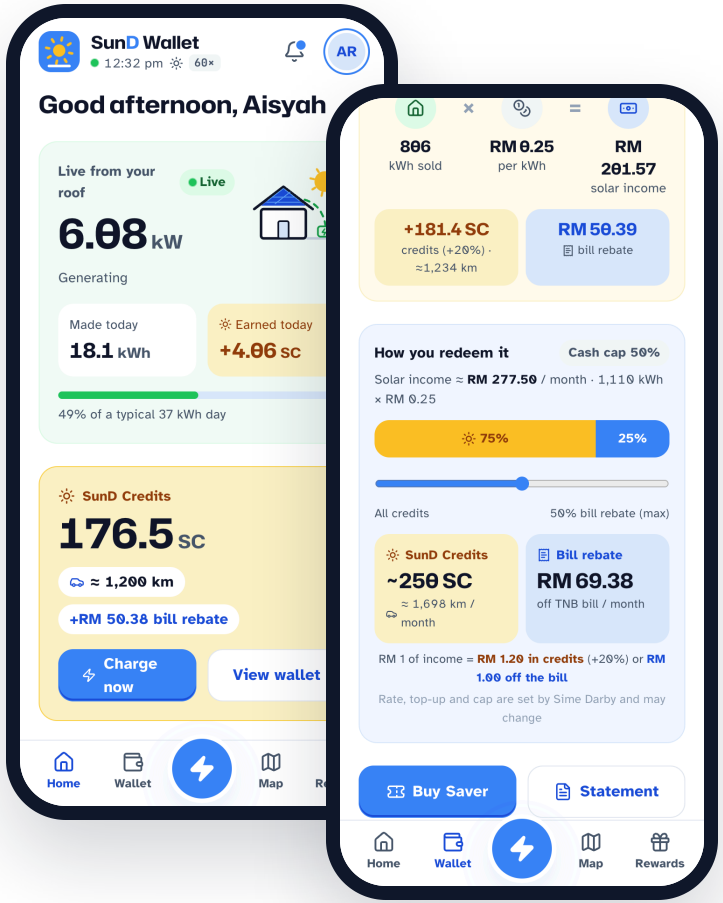
- Solar, but no link to EV demand
- **New revenue**, ESG, a reason to buy SD

Beyond drivers: SunD eBike, Gym and Clubhouse bring in kids, families and seniors too

Scenes from the SunD concept illustrations (Appendix A1, A2)

SunD

SunD Wallet: owners earn, everyone saves



SD controls CREAM energy and routes it to **nearby commercial areas or data centres**. Owners get rewarded.

1,100 kWh
sold / month

→

× RM 0.25
normal selling rate

→

≈ RM 275 / month

SunD Credits +20% · unlimited — Bill rebate · ≤50%

always ≥ 50% | owner's choice

Owner slides the split · rebate never above 50%

RM 1 of income as credits
RM 1.20
to spend at SunD

vs

RM 1 of income as rebate
RM 1.00
off the TNB bill

Credits top-up
+20%
cheap for SD: bay solar ~15 sen/kWh

Owner's choice
EV owner? Go all credits. No EV yet? Take up to half off the bill.

Figures SD can tune
RM 0.25 rate, +20% top-up, 50% cap: all adjustable later.

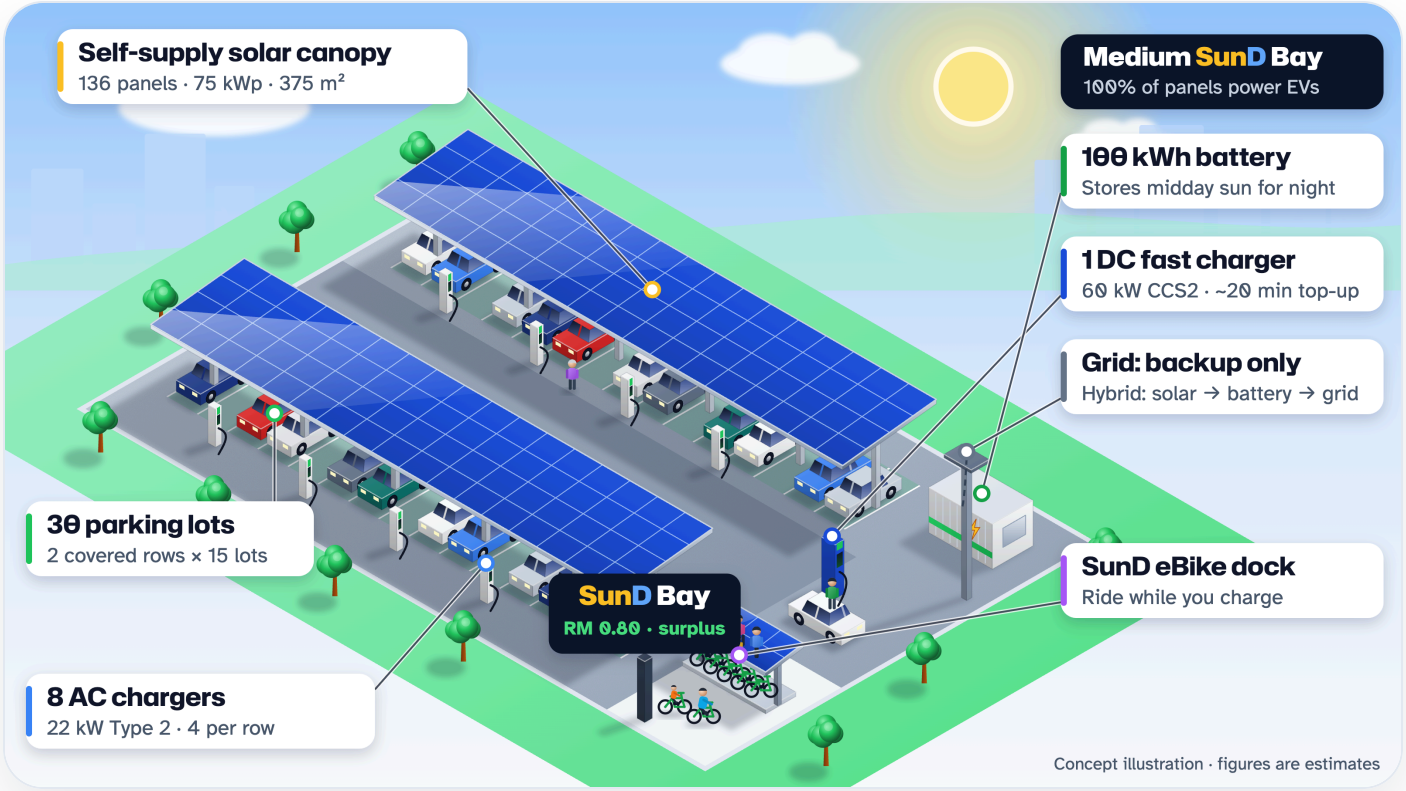
Public: Saver Credits
Pay RM 20, get RM 30. Use at high-solar bays and SunD eBike.

Estimates: 10 kWp roof ≈ 1,100 kWh/month · RM 0.25/kWh benchmark: avg system marginal price ~RM 0.20–0.25.

Live prototype screens: Home and the redemption mix in Wallet

SunD

SunD Bay: solar-first hybrid charging



Independent of CREAM: canopy → battery → grid backup, so a bay works anywhere

Open to all, and cheaper

~RM 1.00 DC vs RM 1.20–1.80 today · or pay with SunD Credits

Solar Surplus Discount

Battery ≥80% → price drops automatically, nothing wasted

Saver Credits: RM 20 → RM 30

Public prepay, valid only at high-solar bays

Bonus credits +5–10%

For choosing SunD over fossil-grid bays

Inside	SD malls · parks · clubhouses · condo car parks
Beyond	Other townships · offices · highways · petrol stations
Modular	Add, move or resize canopies by demand

Concept illustration: Medium SunD Bay · full view in Appendix A1 · figures are estimates

SunD

Not another charging network: a closed solar loop

	Today	With SunD
Reward	All cash, leaks out	**Income: credits +20% or ≤50% bill rebate**
Energy path	Shared solar: 5 km cap + grid charges	**On-site canopy: 0 km, no charges**
Reach	Township-only perks	**Open to all, scales beyond SD**
Facilities	Charging only	**Credits pay for eBike, Gym, Clubhouse**
Midday solar	Wasted	**Stored + Surplus Discount**
Power	Grid bays, ~92% fossil	**Solar first, grid backup only**
Build	Fixed sites	**Modular, demand-based**

OUR EDGE

Turns CREAM rooftop value into SunD Credits, while SunD Bays run on their own sun, anywhere.



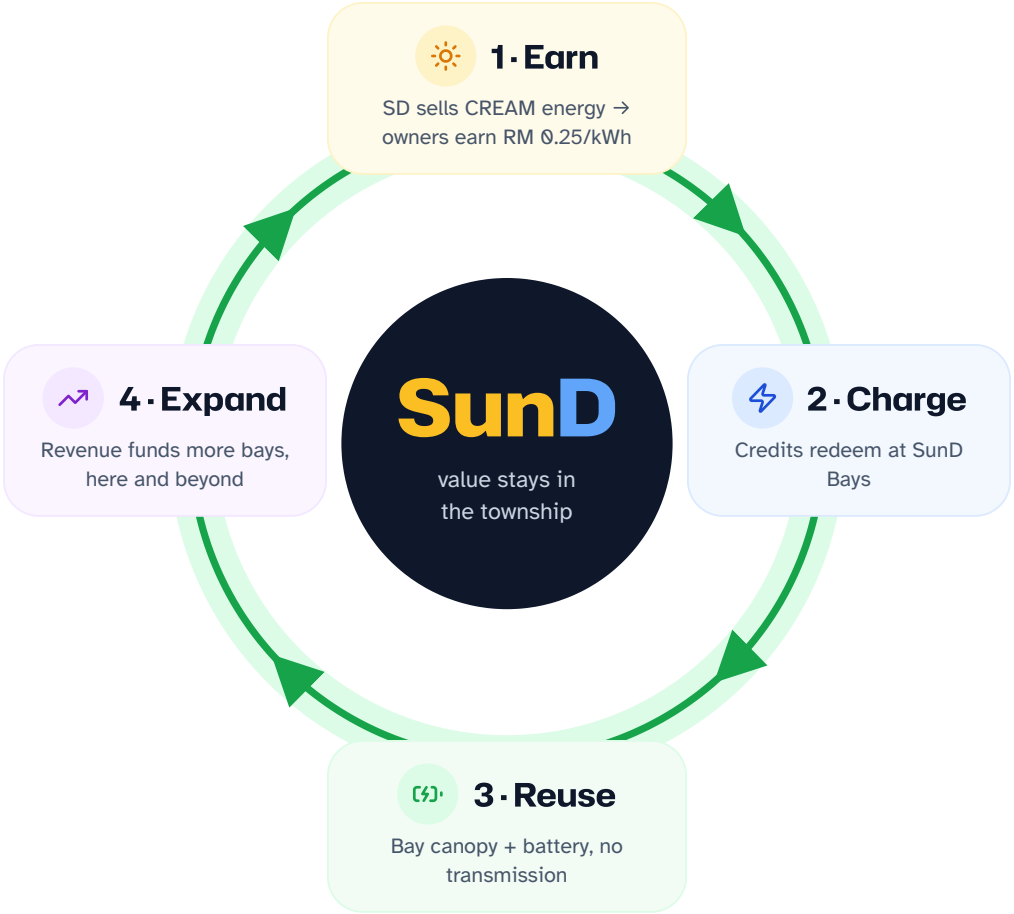
One wallet, many green places

Bays, eBike trail, Gym, Clubhouse · public welcome

CAC: CREAM Community Access Charge · CRESS: Corporate Renewable Energy Supply Scheme (Oct 2026 rates)

SunD

One loop: earn → charge → reuse → expand



TWO ENGINES, ONE WALLET

CREAM engine

SD routes roof energy to commercial areas or data centres. Owners earn RM 0.25/kWh sold.

SunD Bay engine

Own canopy + battery charges any driver, at any site.

WHAT KEEPS IT SPINNING

Battery sets the price

≥80% surplus discount · ≥60% Saver OK · <15% grid backup

Public pays less in the sun

Surplus ~RM 0.80 · Saver +50% · map points to the sunniest bay

Bonus, streaks, eBikes, clubhouse

More reasons to spend credits on green



Energy efficiency · Green transport · Circular economy · Climate resilience

SunD

Every side wins, the planet included

Public

~RM 1.00 DC vs RM 1.20–1.80 · surplus ~RM 0.80 · Saver +50%

More people switch to EVs

SD owners

Solar income ~RM 275/month

As credits (+20%) or bill rebate

Sime Darby

CREAM energy to its commercial sites, data centres

New revenue · ESG: Scope 1&2 –40% by 2030

Planet

~40% of bay energy from on-site solar

Clean charging for everyone

14 km

of driving a day from **1 solar panel**

250 km

a day from **1 SD roof**, ≈ 2× around Earth a year

112k km

a day from **450 Elmina homes** (112,000 km), ≈ 2.8× around Earth

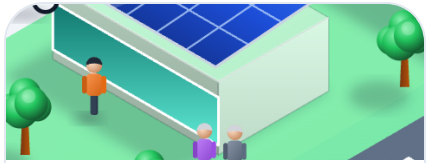
–43%

energy cost per bay: **RM 95 → RM 54** a day



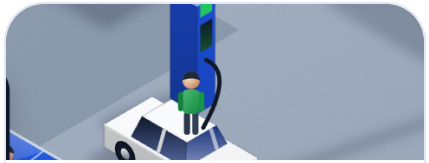
Kids & parents

SunD eBike rides



Seniors

Solar SunD Gym



Public drivers

Cheaper, Saver +50%



Families

Clubhouse + lake park

Reach grows

Residents → public → other townships → nationwide

All figures are estimates · 1 panel ≈ 2 kWh/day · 6.8 km/kWh · SD ESG targets from Sime Darby Property.

SunD

Realistic today: SD already owns the hardest pieces

Fair comparison · same site: 10 lots, 1 DC + 3 AC, ~210 kWh/day demand

	Existing bay	SunD Bay
Source	Grid, ~92% fossil	**Canopy 25 kWp + battery**
Solar share	0%	**~40%**
Price DC / AC	RM 1.20–1.80 / 0.75–1.15	**~RM 1.00 / ~0.70**
Energy cost / day	~RM 95	**~RM 54**
Peak demand	Full	**Cut by battery**
Setup cost	RM 70k–105k	**RM 220k–380k**

Honest trade-off: higher upfront cost, matched by ~35% more usage, which SD credit users bring.

Already in place

- CREAM Elmina: 4.5 MW on 450 homes, live since Sep 2025
- **CREAM is brand new for Malaysian landed homes**, the ideal moment to build SunD into future SD developments
- SD-owned malls, parks and clubhouses as bay sites

Stakeholders

SD Property (owner) · TNB / GSPARX (CREAM) · charger + battery vendors · residents and public

4–6 yrs
carport payback

60%
GITA tax allowance

2–3 mo
small bay live

Demand risk: very low · Built-in users: credits spend only here · Cheapest solar charging nearby · Saver pre-sales = cash upfront · Modular: move quiet canopies

Estimates · costs from 2026 Malaysian vendor ranges (carport RM 4–6k/kWp, battery RM 1–2.5k/kWh, DC 60 kW RM 60–80k).

SunD

From one Elmina pilot to a SunD sub-brand

PHASE 1 · 0–6 MONTHS
Pilot

SunD Wallet for Elmina CREAM homes

1 Small + 1 Medium bay · design SunD into new SD launches

PHASE 2 · 6–18 MONTHS
Township

Expand bays where usage data says so

Open to public · launch Saver Credits

PHASE 3 · 18–36 MONTHS
Beyond SD

SunD Bay sub-brand: other SD townships, malls, highway stops, petrol stations

Same wallet, every location

PHASE 4 · 3+ YEARS
Platform

SunD eBike, gym, clubhouse · EV fleets

License SunD to other developers and councils

Not limited to the township

One SunD Wallet · every SunD location · every EV driver

Elmina pilot
2 bays · 450 homes

SD townships
6,000 rooftops

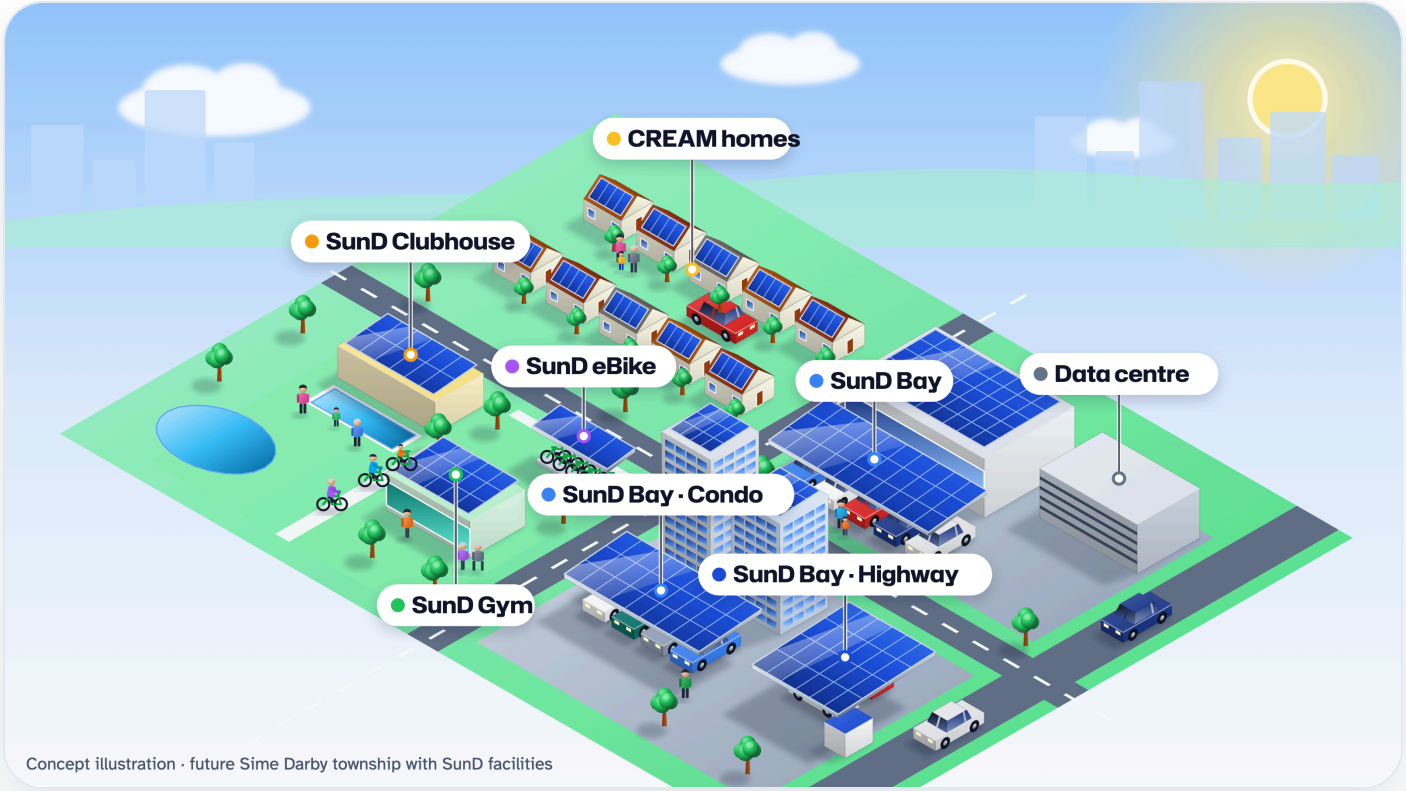
Malls · offices · campuses
any site, own canopy

Highways · petrol stations
charge on the go

Other developers · councils
license SunD

Pilot success = **≥40%** solar share per bay · Utilisation **well above ~1%** national · **Credits redeemed** vs issued · **Non-SD buyers** citing SunD

Life in a SunD township, ready to pilot



Concept illustration · future Sime Darby township with SunD facilities

Working SunD Wallet prototype: screens in Solution 1 and Appendix A3

Why SunD deserves to win

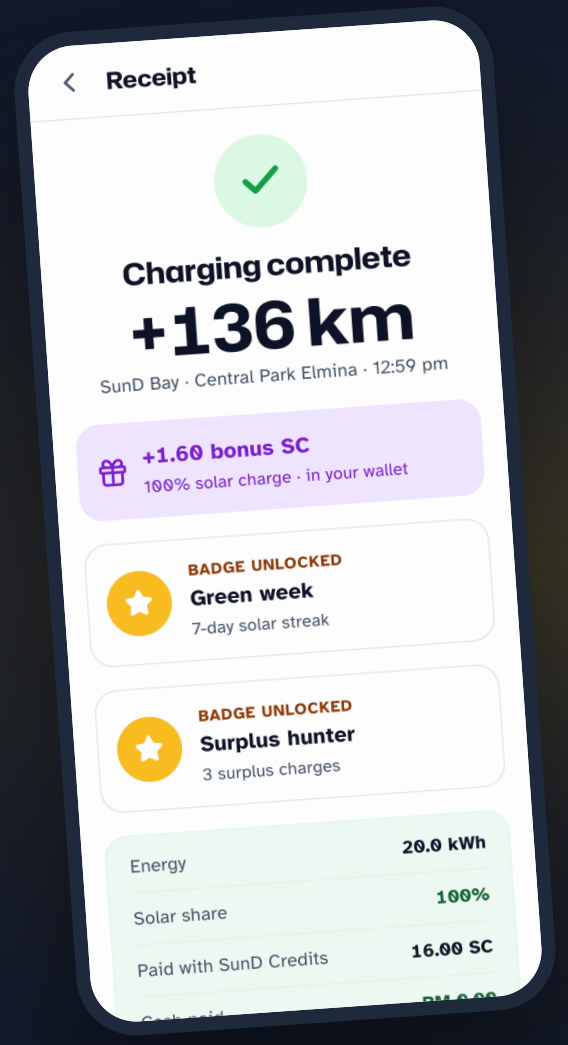
SunD turns Sime Darby's new CREAM rooftops into a closed solar loop that benefits everyone. Owners earn income on every kWh sold and choose credits (+20%) or a bill rebate. The public charges cheaper, with Saver Credits and surplus discounts that turn EV interest into EV ownership. Credits also unlock SunD eBike, Gym and Clubhouse, so demand is built in. Bays run on their own sun, inside and beyond townships. It is feasible on SD's own sites, de-risked by pre-sold credits and modular canopies, and already a working prototype.

- Live prototype
- Cheaper for the public
- Low demand risk
- Beyond SD and beyond charging

SunD

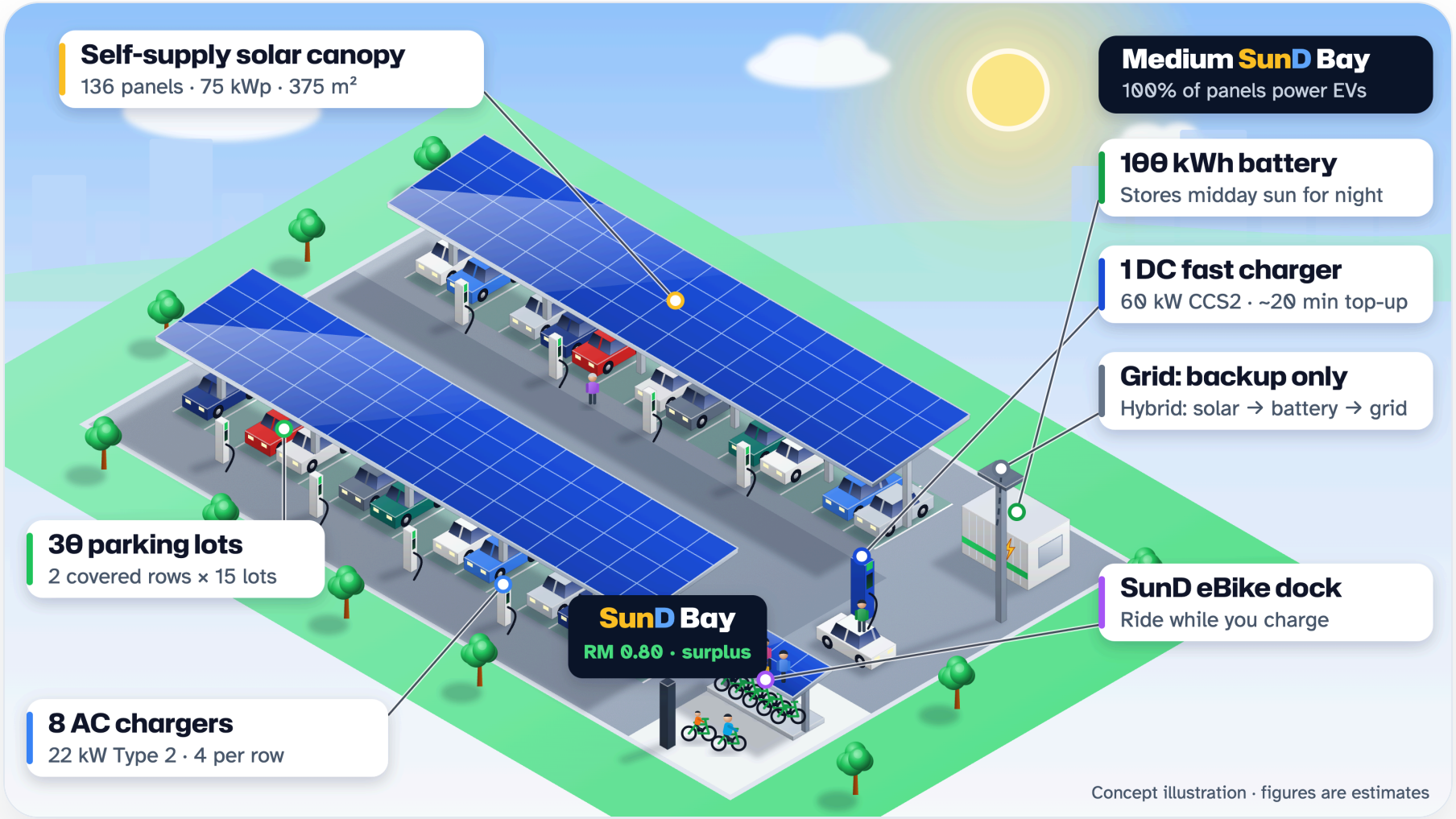
Your roof earns it.
Your car drives it.
Your township lives it.

Thank you · Team UrbanByte · Appendix follows →



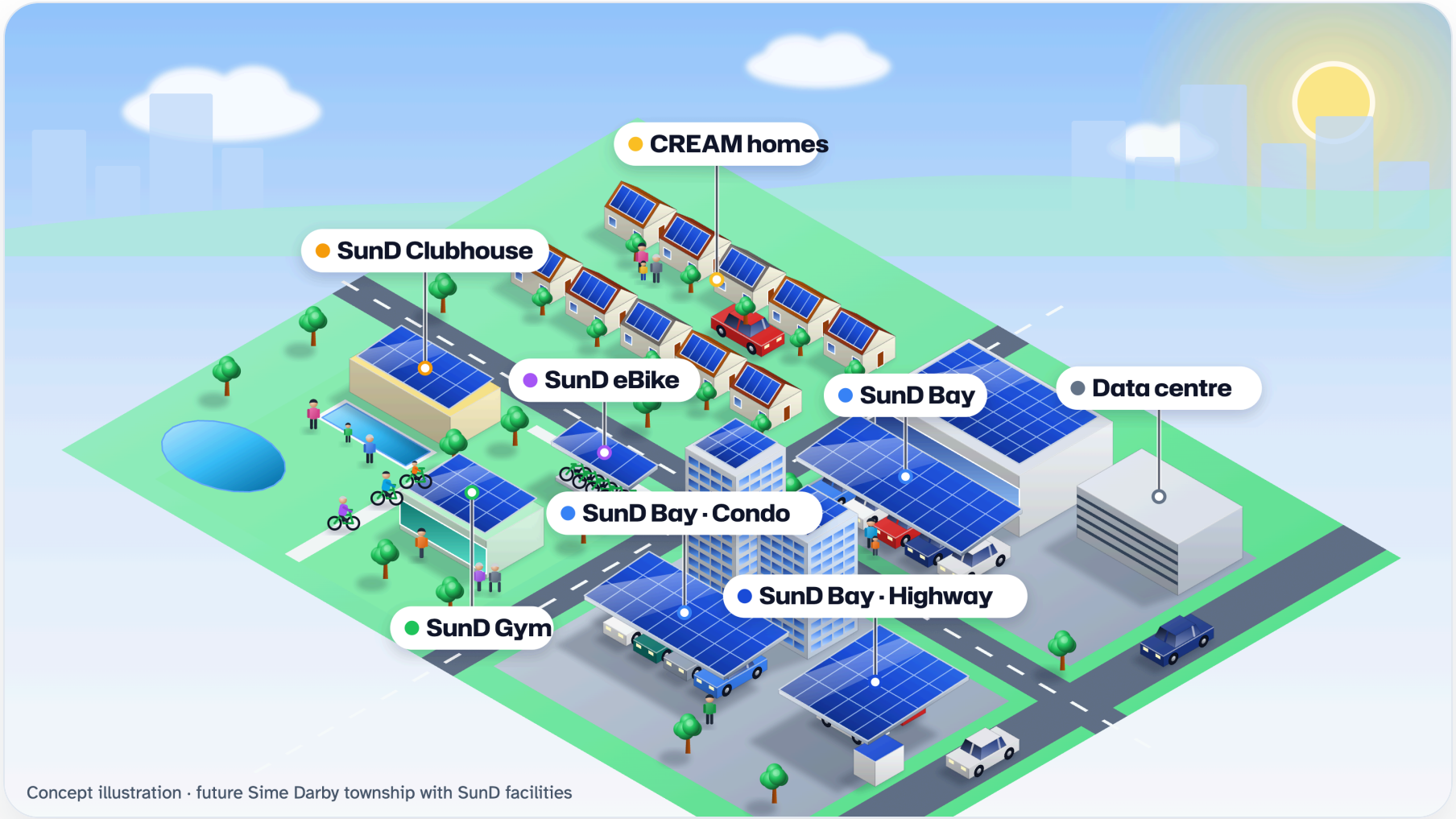
SunD

SunD Bay on site: a Medium bay, labelled



SunD

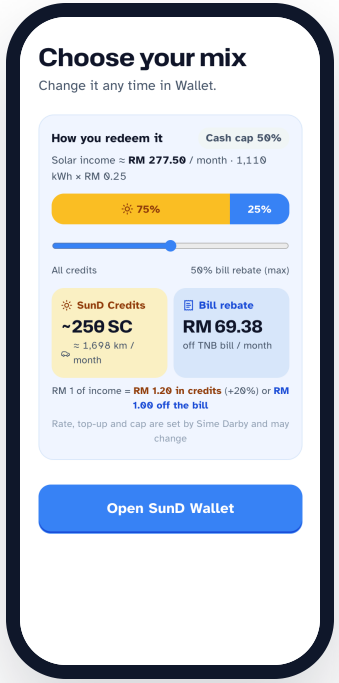
Life in a future SunD township



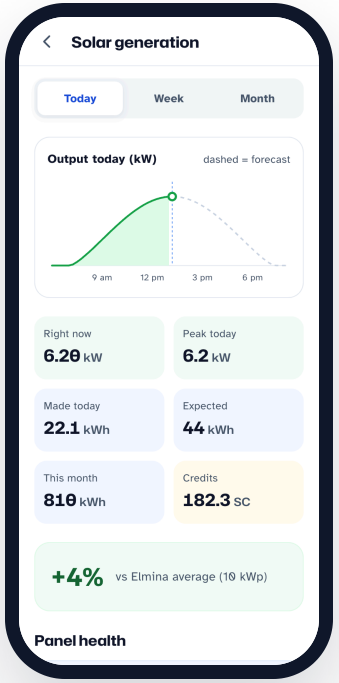
Concept illustration · future Sime Darby township with SunD facilities

SunD

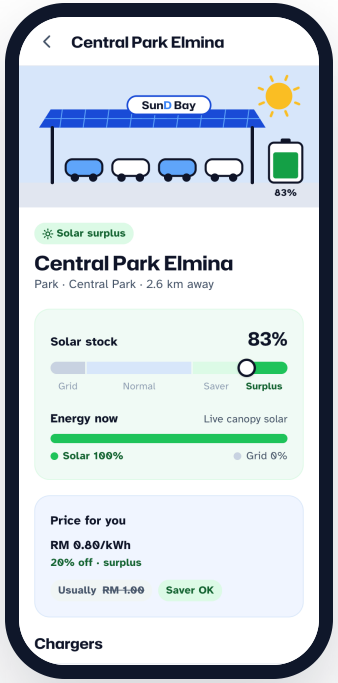
More SunD Wallet screens



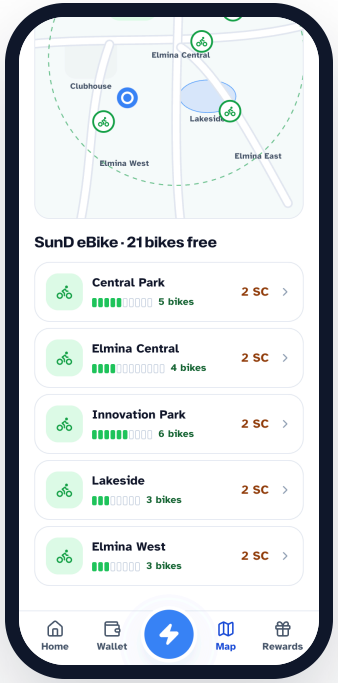
Reward mix
Credits ≥50% · rebate ≤50%



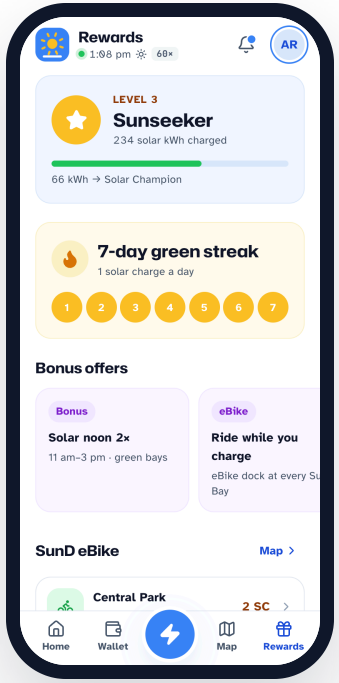
Solar detail
Live output curve



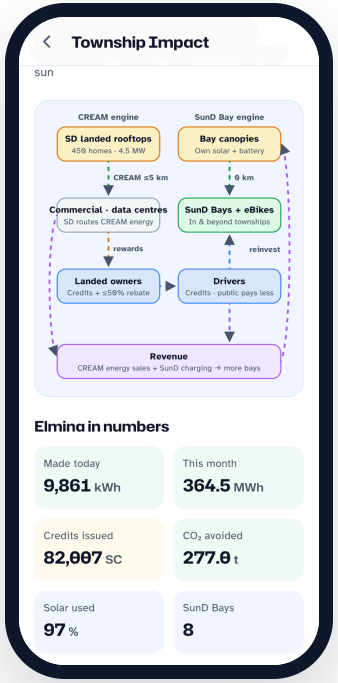
Public view
Surplus RM 0.80



SunD eBike
Live dock slots



Rewards
Level, streak, offers



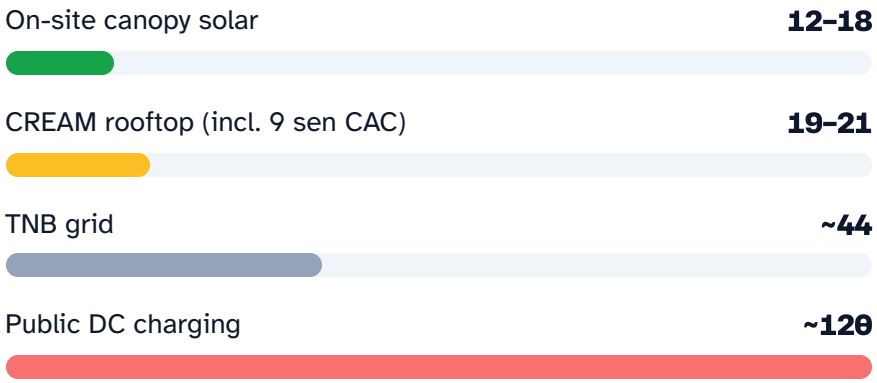
Township Impact
Energy loop

SunD

Solar math: panels to kilometres, and cost per kWh

Panels	Where	kWh/day	km/day	Cars (50 km top-up)
1	Single panel	2	14	¼
4	One commuter	8	54	1
19	**1 SD roof**	37	250	5
45	Small bay (10 lots)	90	610	12
136	Medium bay (30 lots)	270	1,840	36
455	Large bay (100 lots)	900	6,100	120
8,550	**450 Elmina homes**	16,500	112,000	2,200

Cost per kWh (sen)



On-site canopy avoids the 9 sen CAC and transmission loss.

Estimates: 1 kWp ≈ 110–130 kWh/month · 550 W panel ≈ 2 kWh/day · e.MAS 5 ≈ 6.8 km/kWh · TNB domestic ≈ 44.4 sen/kWh.

SunD

SunD Bay sizes, costs and setup time

Small

10 lots

25 kWp · 4 AC · 50 kWh

RM 175k–325k

2–3 months · clubhouse, park

Medium

30 lots

75 kWp · 8 AC + 1 DC · 100 kWh

RM 510k–880k

3–4 months · mall, commercial

Large

100 lots

250 kWp · 16 AC + 2 DC · 300 kWh

RM 1.5M–2.6M

4–6 months · highway, flagship

Unit costs

Carport solar **RM 4–6k / kWp**

AC charger **RM 3–8k**

DC 60 kW **RM 60–80k**

Battery **RM 1–2.5k / kWh**

DC meter **4–12 weeks**

Estimates · trexon.my 2026 ranges · 60% GITA applies to charger installation · carport payback ~4–6 years.

SunD

Where SunD prices sit in today's market

Network	DC (RM/kWh)	AC (RM/kWh)
Brand Y	0.85–1.10	from 0.40
Brand J	1.20–1.40	from 0.75
Brand G	1.30–1.80	0.90–1.05
Brand D	1.00–1.50	–
Brand S	1.20–1.80	–
Brand T (other EVs)	1.80	–
SunD Bay (proposed)	**~1.00 · or SunD Credits**	**~0.70**

Market signals

- Top 3 networks hold >40% of charge points
- No government price cap (MITI, Jul 2026)
- National utilisation ~1% (peak ~3%)
- Target: 30,000 chargers by 2030

Why SunD fills bays

A built-in user base of SD credit holders lifts utilisation from day one, which is the main weakness of today's networks.

Rates vary by site (evsifu.com, 2026)

SunD

Sources

1. Sime Darby Property: CREAM residential solar launch (simedarbyproperty.com)
2. IEA-PVPS Malaysia 2025: CREAM 5 km, 9 sen CAC, 6,000 rooftops
3. TNB analyst briefing 4QFY23: 450 homes, 4.5 MW
4. carz.com.my (Sep 2026): EV registrations vs chargers
5. paultan.org / MITI (Jul 2026): 6,416 public chargers, 30,000 by 2030
6. EY Malaysia EV consumer survey (Oct 2024)
7. The Edge / Reuters: Peninsular power mix 2025
8. The Star (Oct 2026): CRESS charges
9. evsifu.com: EV charging network rates 2026
10. S&P Global Mobility: network utilisation
11. trexon.my: solar carport and EV charger installation costs 2026
12. wapcar.my: Proton e.MAS 5 specifications